SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Computer Vision with OpenCV
COURSE CODE: DSA0201

COURSE OUTCOME COVERED

CO2: *Apply image processing and feature extraction techniques, including edge detection, motion estimation, and optical flow, for analyzing images.. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks:50**

Annexure A

SCENARIO BASED

Code of Conduct:

I ***B. Deepthi (192424269)*** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B.Deepthi

Question 1: Crack Detection on Concrete Surfaces Using Drone Images

A. Image Preprocessing Pipeline

Step 1: Illumination Correction

- Apply CLAHE (Contrast Limited Adaptive Histogram Equalization) or background illumination correction.
- **Reason:** Removes the effect of uneven sunlight and shadows, making crack regions uniformly visible.

Step 2: Noise Reduction

- Use a **Gaussian Filter** (or Median Filter if salt-and-pepper noise exists).
- **Reason:** Eliminates Gaussian noise caused by drone vibration while preserving important crack information.

Step 3: Contrast Enhancement

- Enhance image contrast using **Adaptive Histogram Equalization (CLAHE)**.
- **Reason:** Makes thin cracks darker than the surrounding concrete, improving visibility.

Step 4: Image Normalization

- Normalize pixel values between **0 and 1** (or 0–255).
- **Reason:** Produces consistent image intensity, improving the performance of edge detection algorithms.

Final Pipeline

Drone Image → Illumination Correction → Gaussian Filtering → CLAHE → Normalization → Edge Detection

B. Suitable Edge Detection Technique

Chosen Method: Canny Edge Detector

Why Canny?

- Detects extremely thin cracks accurately.
- Uses Gaussian smoothing to reduce noise.
- Employs double thresholding to remove weak false edges.
- Performs edge tracking, resulting in continuous crack boundaries.

Comparison with Sobel Operator

Canny Edge Detector	Sobel Operator
High accuracy	Moderate accuracy
Removes noise before detection	Sensitive to noise
Detects fine cracks	Misses thin cracks
Produces continuous edges	Produces thick or broken edges

Conclusion

Canny is more suitable because concrete cracks are narrow, irregular, and affected by outdoor lighting conditions.

C. Post-Processing Technique

Morphological Closing (Dilation + Erosion)

- Connects broken crack segments.
- Fills small gaps.
- Produces continuous crack structures.

Additional Improvement: Skeletonization can be applied to obtain the centerline of the crack for accurate measurement.

Question 2: Human Motion Tracking Using Security Camera

A. Lucas-Kanade Optical Flow (4 Marks)

Lucas-Kanade estimates the movement of feature points between two consecutive video frames.

Working

1. Detect corner features (Shi-Tomasi or Harris).
2. Assume small movement between frames.
3. Calculate displacement using image intensity changes.
4. Generate optical flow vectors showing direction and speed.

Advantages

- Computationally efficient.
- Suitable for real-time surveillance.
- Works well when motion between frames is small.

B. Aperture Effect

The aperture effect occurs when motion is observed through a small image region.

Impact

- Plain walls and smooth floors contain very few features.
- Clothing with uniform colour provides insufficient texture.
- Optical flow cannot determine the true motion direction.
- Motion estimation becomes ambiguous or inaccurate.

Result

Tracking may drift or lose the target.

C. Sudden Speed or Direction Change

When a person suddenly changes direction:

- Optical flow vectors become larger.
- Vector direction changes abruptly.
- Previous motion prediction becomes inaccurate.

Improvement Method

Use a **Kalman Filter** together with Lucas-Kanade.

Benefits

- Predicts future object position.
- Smooths noisy motion vectors.
- Maintains tracking during sudden movements or temporary occlusion.

Question 3: Vehicle Tracking at Traffic Junction

A. Optical Flow for Vehicle Tracking

Optical flow calculates pixel displacement between consecutive frames.

Working

1. Detect moving vehicle features.
2. Compute displacement vectors.
3. Vector direction indicates vehicle movement.
4. Vector length indicates vehicle speed.

Application

- Speed estimation
- Direction detection
- Vehicle trajectory analysis
- Traffic monitoring

B. Two Challenges Affecting Motion Estimation

1. Occlusion

Problem

- Vehicles overlap while crossing.

Effect

- Optical flow loses visible feature points.
- Incorrect tracking or identity switching occurs.

2. Shadows

Problem

- Moving shadows appear similar to moving vehicles.

Effect

- Optical flow detects false motion.
- Tracking accuracy decreases due to incorrect vectors.

C. Improvement Method

DeepSORT with Kalman Filter

Reason

- Maintains vehicle identity during overlap.
- Predicts future position during temporary occlusion.
- Handles sudden turns and speed changes.
- Produces stable and reliable multi-vehicle tracking.